

TABLE III
MEASUREMENT AND POWER SUPPLY TOOLS

Name of tool	Function
Auto-ranging digital multimeter	An auto-ranging digital multimeter is a must-have tool for beginners as well as advanced users. Look for features that suit your purpose well. A good digital multimeter should have a micro-ampere range and large backlit display. It should have separate input sockets for current measurements and voltage/resistance/other measurements, so that you do not accidentally short the test leads while in current range.
Analogue multimeter	Due to provision of measurement over a dial, an analogue multimeter can be used to test such electronic components that we cannot with a digital meter. Although new digital multimeters come with a bar graph display but they still cannot beat analogue multimeters when it comes to cost. Analogue multimeters are not auto-ranging and you would have to select the range manually.
Logic probe	It is used to check logic levels (high or low) of the digital circuit under test. It has two alligator/crocodile clips that are connected to the supply voltages of the equipment that is being tested.
Lab power supply	A lab power supply is used to power electronic circuits/projects. If you work on op-amps a lot, you should consider buying a dual-tracking power supply. It has two knobs: one for limiting the current and the other for adjusting the voltage. Voltage output is provided at binding posts, which is where banana plugs fit in. For most jobs 0V-18V, 3A output is more than adequate.

TABLE IV
PROTOTYPING TOOLS

Name of tool	Function
Large breadboard	It is good to build your own breadboard by sticking three 840-tiepoint breadboards on a wooden board. Using a drill attach two binding posts (red and black) on the wooden board that will receive power from the lab supply.
Connecting jumper wires	You can either buy pre-made jumper wires or make your own using a single-strand 22SWG hook-up wire. Such wires can be used to add a connection between electronic components on a breadboard or perfboard.
Perfboard	Called perfboard, veroboard or general-purpose PCBs, these serve as a base upon which electronic components are interconnected with each other, so that these can communicate and form a circuit. When buying one, look for the tinned version.

circuit board (PCB). Leads of electronic components are joined to the PCB by melting the solder wire and waiting for it to cool and solidify. PCBs either have external copper wires (in general PCBs/perboards) or printed copper tracks (advanced FR4-grade PCBs) that help in establishing the required connections. Various soldering tools are described in Table I.

Hand tools

Hand tools are required for stripping wires, cutting component leads, drilling holes in PCBs and cabinets, shaping electronic enclosures, setting trimmers and presets, opening enclosures, cutting cabinets, providing assistance in soldering, extracting integrated circuits (ICs), holding things while soldering or other work, magnifying work area and so on. Check out Table II for some basic hand tools.

Measurement and power supply tools

Measurement tools are required in taking measurements of physical quantities in electronics like voltage, current and resistance, while power supply is used to power up the electronic circuits. Table III shows some handy measurement and power supply tools.

Prototyping tools

Prototyping tools are required for assembling a single unit of electronic project/circuit, which then can be mass produced, if successful. Some common prototyping tools are shown in Table IV.

Screwdriver tools

These are your best friends when it comes to opening/hacking electronic gadgets. These come in various tip types like slotted, Philips, Torx and so on. Avoid the extremely-cheap, low-quality screwdriver sets whose tips bend while opening a screw. Select sets that have tweezers included, as these are extremely useful in holding fine wires while soldering. Fig. 2 shows some widely-used screwdriver tools. **EFY**

Fig. 2: Some common screwdriver tools